NASCON'26 · BUILD A PERSONAL MORNING PAPER (EYNVISION)

ParhLo

Architecture and system design

One short, personal, checked newspaper every morning: on the web, as a PDF and in your inbox. It learns from what you read and updates within seconds when your interests change.

1 System architecture

2 Daily pipeline: news in, paper out

3 Instant reprint when interests change

4 How ParhLo learns and gets better every day

5 Ask your paper and search

6 Data model (Neon Postgres + pgvector)

7 Morning email queue and CI/CD

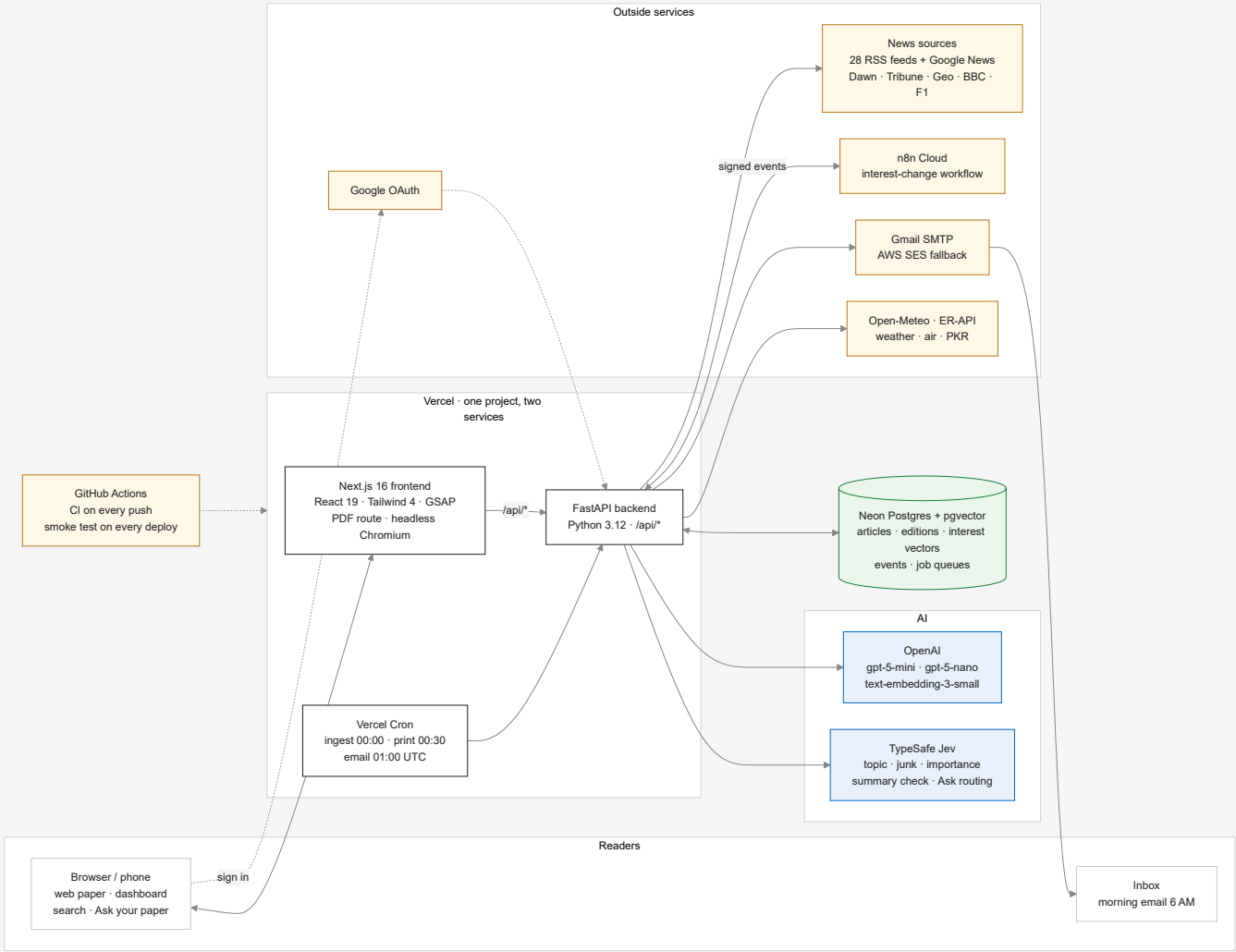
Live parhlo-brown.vercel.app

Code github.com/Tehman700/ParhLo

Stack Next.js · FastAPI · Neon Postgres + pgvector · OpenAI gpt-5 · TypeSafe Jev · n8n · Vercel

1 · System architecture

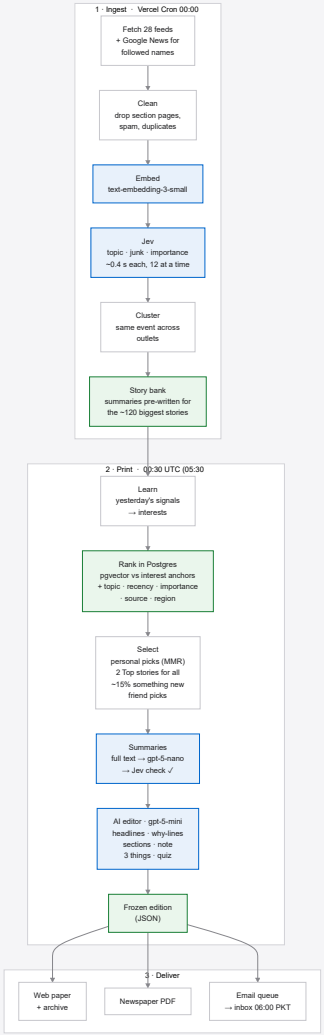
One Vercel project runs the Next.js frontend and the FastAPI backend. Postgres with pgvector on Neon holds everything, including the interest vectors and job queues. OpenAI writes and edits; Jev, a fast decision model, classifies stories, checks summaries and routes questions. n8n orchestrates paper updates; GitHub Actions checks every push and every deploy.



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2 · Daily pipeline: news in, paper out

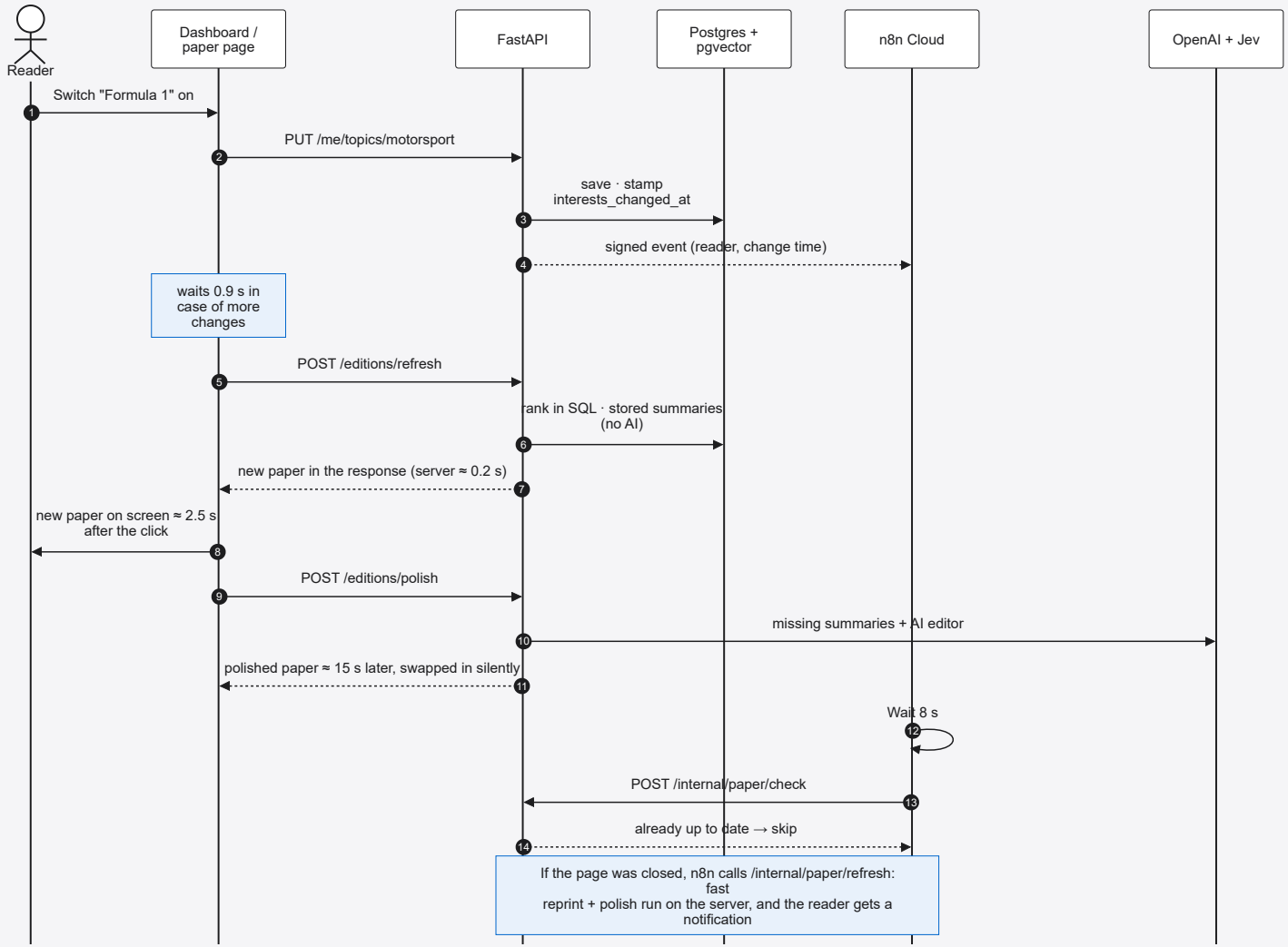
Every night: fetch and clean the news, embed it, let Jev decide topic, junk and importance, merge the same event across outlets, and pre-write summaries for the biggest stories. Then, for every reader, learn from yesterday, rank inside Postgres, select a balanced paper, summarize and check, edit, and freeze one edition for the web, the PDF and the morning email.



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3 · Instant reprint when interests change

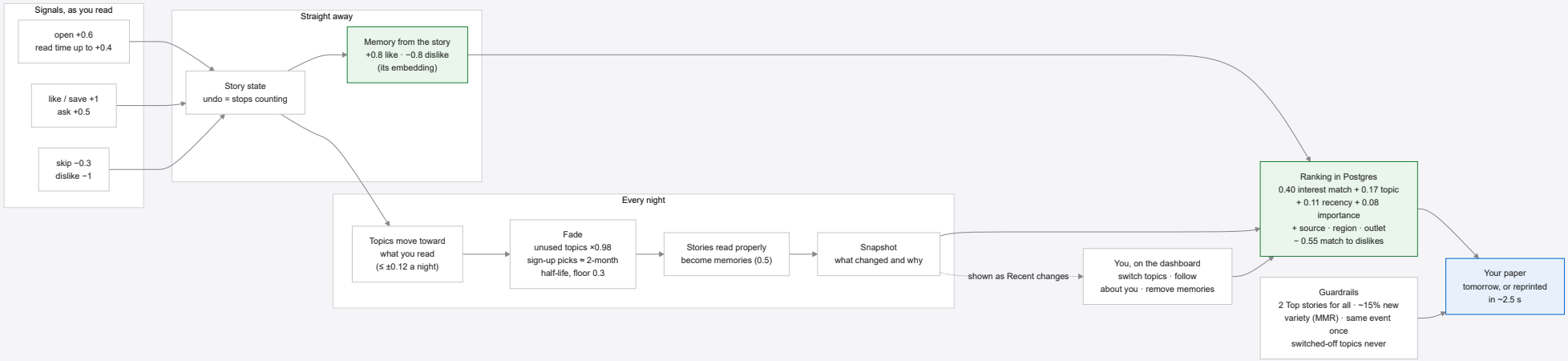
The page asks for a fast print (no AI in the critical path) and shows the new paper about 2.5 s after the click. AI polish follows in the background. n8n receives a signed event for every change and guarantees the reprint when the page is closed; when the page got there first, its check says 'already up to date'.



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4 · How ParhLo learns and gets better every day

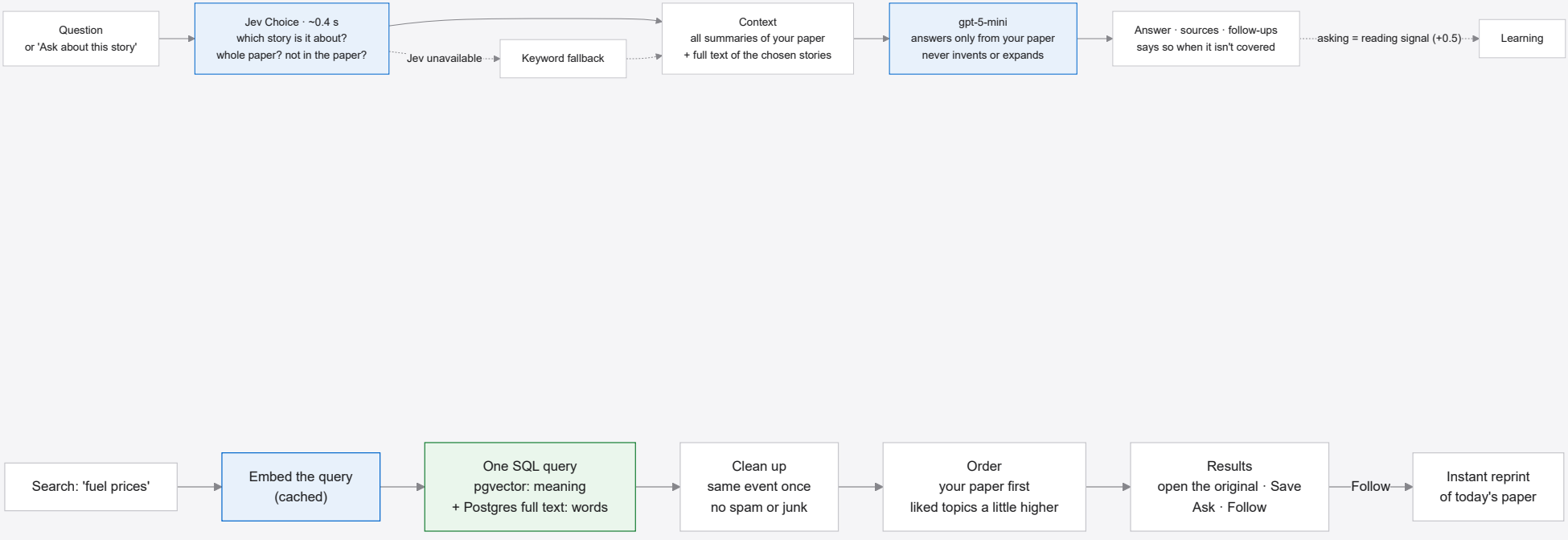
Every action is a signal. Likes and dislikes become memories at once; every night topics move toward what you read, unused interests fade, and sign-up picks give way to real reading. Everything is visible and editable, and guardrails keep the paper from becoming a bubble.



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5 · Ask your paper and search

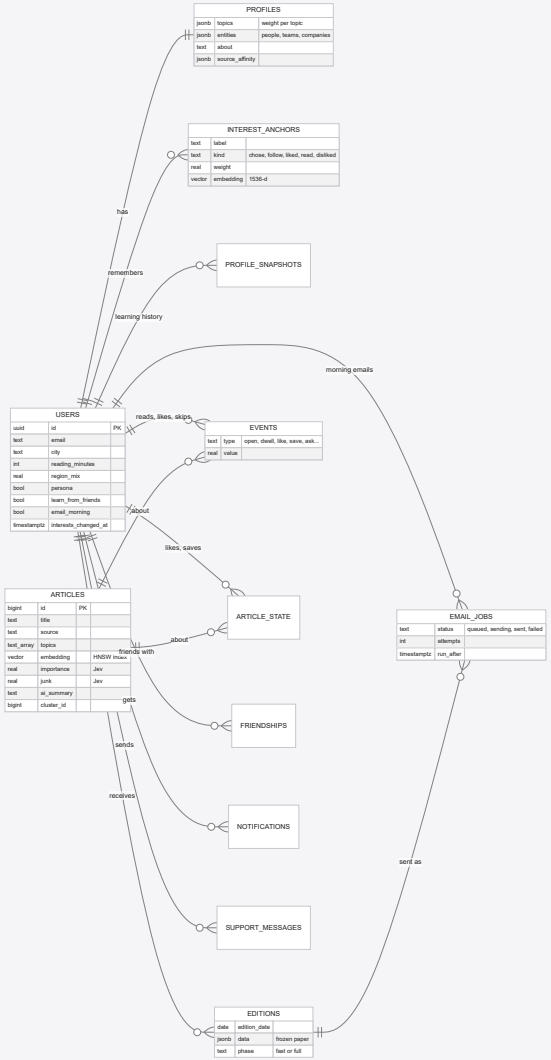
Top: Ask your paper. Jev picks the story a question is about, then gpt-5-mini answers only from the reader's own paper, with sources. Bottom: search by meaning and by words in a single SQL query, with the reader's paper first; Follow turns a search into an interest and reprints the paper.



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6 · Data model (Neon Postgres + pgvector)

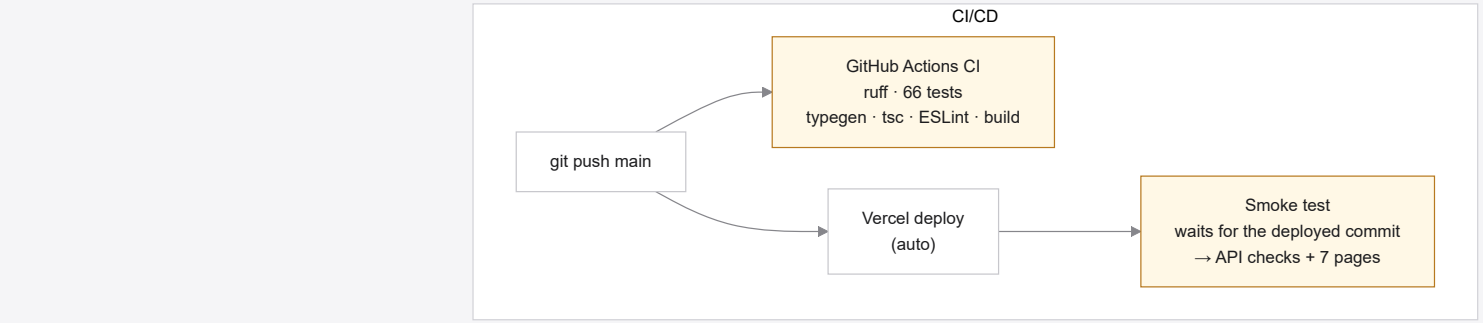
Readers, their interests (topic weights plus vector memories), the frozen daily editions, every reading event, likes and saves, friends, notifications, the email job queue and help messages. Articles carry an embedding (HNSW index) plus Jev's importance and junk scores.

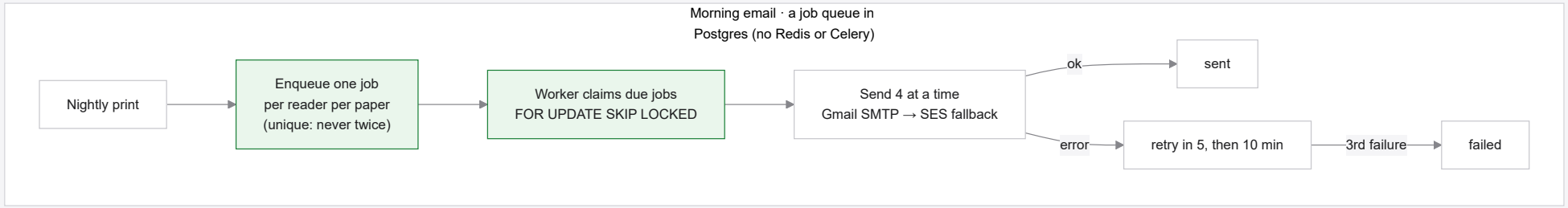


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7 · Morning email queue and CI/CD

The morning email is a job queue in Postgres: idempotent, claimed with SKIP LOCKED, retried with backoff. Every push runs CI (lint, 66 tests, type check, build); every production deploy runs a live smoke test that waits for the deployed commit.





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